SURVEYING & LEVELLING · NOTE SL11

Differential, Profile and Cross-Section Levelling — Field Procedure and Booking

Course topics: SWCE1.4 L10, SWCE1.4 L11

Setting up and levelling the instrument

1. Spread the tripod legs about 1 m apart, press them firmly into the ground and keep the tripod head roughly level.
2. Place the telescope over two levelling screws and centre the bubble by turning both screws simultaneously.
3. Rotate the telescope 90° and centre the bubble with the third screw; repeat until the bubble stays central in every direction.
4. Focus the eye-piece on the cross hairs, then the telescope on the staff — this removes **parallax** (apparent movement of the hairs when the eye moves).
5. Check the bubble before every reading and read the staff at the middle horizontal hair.

Differential levelling

Differential levelling finds the difference in elevation between two points.

- **Points near each other:** set the level roughly equidistant from both; the difference in level = difference of the rod readings.
- **Points far apart:** make a series of set-ups along the route, using turning points.

1. Start on a bench mark of known or assumed elevation.
2. Take a BS (plus sight) on the BM; HI = RL of BM + BS.
3. Take an FS (minus sight) on a turning point; RL of TP = HI − FS.
4. Shift the level, take a BS on the TP and continue until you reach the final BM.
5. Run a closing line back to the start as a check.

Booking: two systems

System	How it works	Arithmetic check
Collimation (height of instrument)	Compute the HI for each set-up; RL = HI − reading	ΣBS − ΣFS = last RL − first RL
Rise and fall	Compare each reading with the previous one: a smaller reading is a **rise**, a larger one a **fall**	ΣBS − ΣFS = ΣRise − ΣFall = last RL − first RL

Worked example. BM (RL 100.000 m): BS 1.525; P1: IS 1.850; TP1: FS 2.105, BS 0.985; P2: IS 1.420; P3: FS 1.760.
Collimation: HI_1 = 101.525 → RL P1 = 99.675, RL TP1 = 99.420; HI_2 = 99.420 + 0.985 = 100.405 → RL P2 = 98.985, RL P3 = 98.645.
Rise and fall: falls of 0.325, 0.255, 0.435 and 0.340 give the same RLs.
Check: ΣBS − ΣFS = 2.510 − 3.865 = −1.355 = ΣRise − ΣFall = 98.645 − 100.000.

Profile (longitudinal) levelling

Profile levelling, or sectioning, finds the elevations of points at measured intervals along a chosen line. Plot the elevations on profile paper to study grades, depths and high and low spots and to estimate cuts and fills — essential for roads, ditches, terrace lines, contour bunds and tile lines. Take the levels along the centre line of the proposed road, terrace, watercourse, drainage channel or structure.

1. Locate and station the profile line before levelling; set a BM near the start whenever possible; peg the points where elevations are needed.
2. Set up where several profile stations are visible. Take the first reading (a plus sight, BS) on the BM or a starting point of known elevation; HI = elevation + BS.
3. Read the rod on every visible station — at regular intervals, at all full stations and at every major break in slope. These are minus sights (IS or FS); elevation = HI - rod reading.
4. When the rodman has moved about 90 m, select a TP, take an FS on it, shift the level, take a BS on the TP and compute the new HI.
5. Drive a hub stake (5 cm × 5 cm × 40 cm) flush with the ground about every 150 m or closer.
6. Always close the circuit back to the starting BM or the nearest BM.

- **Error of closure** for a profile uses only the BS and FS on bench marks and turning points — intermediate sights do not enter it.
- **Field book:** sketch changes of direction and nearby landmarks; read the rod at every important point — field channels, tile laterals, gullies, culverts, bridges, roads, fence lines and major breaks in slope. Reduce the levels by either system.

Plotting the profile

1. Draw a horizontal datum line and mark the station distances along it to a convenient scale.
2. Erect perpendiculars at each chainage and set off the elevations on them.
3. Join the plotted points with straight lines to get the ground profile.
4. After designing the work (road, channel…), mark the formation levels; the gaps between ground and formation give the cut and fill heights.

Scales: the horizontal scale follows the station distances — commonly 10 or 20 m to 1 cm; exaggerate the vertical scale, commonly 2 or 3 m to 1 cm.

Other kinds of levelling

Kind	Purpose
Cross-section levelling	Profiles at right angles to a base line (centre line of a road, watercourse or gully); run with the profile levels or after the line has been staked
Check levelling	Re-running levels to check those already fixed; close the lines back to the day's starting point
Flying levels	A quick preliminary run of approximate levels at representative points, to choose the line for detailed levelling
Bench level circuit	Fixes the relative elevations of bench marks; always close it on the starting point; it underpins profile, cross-section and topographic surveys

Hand signals between levelman and rodman

Move in the direction shown · plumb the rod · set a turning point · use the long rod · observation completed · wrong face / check the clamp · move up / move down · come in.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. (Lesson 10) Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. In the rise-and-fall system, a reading smaller than the previous one indicates a

(a) fall
(b) rise
(c) turning point
(d) closing error

Q2. Arithmetic check of the collimation system is

(a) ΣBS − ΣFS = last RL − first RL
(b) ΣRise = ΣFall
(c) ΣIS = ΣFS
(d) ΣBS = ΣFS

Q3. In profile levelling, a turning point is normally chosen after the rodman moves about

(a) 15 m
(b) 30 m
(c) 90 m
(d) 300 m

Q4. Intermediate sights are used in computing the error of closure

(a) always
(b) never
(c) only in rise and fall
(d) only for profiles

Q5. Levels run quickly at representative points to choose a route are

(a) check levels
(b) flying levels
(c) reciprocal levels
(d) profile levels

Answers

Q1 — (b) rise
Q2 — (a) ΣBS − ΣFS = last RL − first RL
Q3 — (c) 90 m
Q4 — (b) never
Q5 — (b) flying levels